Conducting an investigation; taking a sample

From a question to a sample that can honestly answer it. [Cambridge insert]

LESSON 158–159 | 2 × 40 MIN | MATEMATIKA 6, QO‘SHIMCHA MAVZU | S7 6.1–6.2 INVESTIGATIONS

LESSON CLOCK

20' | 20' | 25' | 15'

The statistical cycle	20 min
Asking a good question	20 min
Choosing a sample	25 min
Sources of bias	15 min

LEARNING OBJECTIVES

- Name the five stages of a statistical investigation.
- Write a survey question with a small, fixed list of answers.
- Choose a sample that represents the population it is drawn from.
- Recognise the four common sources of bias and remove them.

TEXTBOOK

National	pp. 453–457
Cambridge	Stage 7, pp. 58–66
Workbook	Exercise 6.1–6.2

01 Explanation

The statistical cycle

Every investigation runs through the same five stages, in the same order. Skipping one is what makes a survey worthless.

STAGE	THE WORK	WHAT IT PRODUCES
1 — plan	decide the question and the answers	a survey question
2 — sample	decide who is asked, and how many	a list of people
3 — collect	ask, and record on a sheet	a tally
4 — process	frequencies, averages, a chart	a table and a diagram
5 — interpret	write what the data says	conclusions, with limits

THE QUESTION COMES FIRST, ALWAYS

Data collected before the question is decided almost never answers it. Five minutes of planning saves an afternoon of asking the wrong people the wrong thing.

Asking a good question

POOR QUESTION	WHAT IS WRONG	BETTER
What do you think about sport?	the answers cannot be tallied	Which of these six sports do you prefer?
Don’t you agree football is best?	it leads the answer	Which sport do you prefer?
Do you do a lot of sport?	“a lot” means different things	How many hours of sport a week?
How old are you and what sport?	two questions in one	ask them separately

OVERLAPPING OPTIONS ARE AS BAD AS MISSING ONES

Age boxes of *10–12*, *12–14* leave a twelve-year-old with two homes and no rule for choosing. Write *10–11*, *12–13*, *14–15* instead — no gaps, no overlaps.

Choosing a sample

The **population** is everyone the question is about; the **sample** is the part actually asked. A sample of *60* from a school of *600* is one pupil in ten.

GRADE	PUPILS	IN A 1-IN-10 SAMPLE
6	*120*	*12*
7	*110*	*11*
8	*130*	*13*
9	*120*	*12*
10	*60*	*6*
11	*60*	*6*
total	*600*	*60*

GIVE EVERY PART OF THE POPULATION ITS SHARE

Each grade contributes a tenth of itself, so the sample looks like the school. Taking all *60* from Grade 6 would give a sample of *60* that answers a different question entirely.

Two honest ways to choose the individuals: draw names from a hat (**random**), or take every tenth name from an alphabetical list (**systematic**). Both are decided before anyone is asked.

Sources of bias

WHAT WAS DONE	WHY THE RESULT IS BIASED	THE FIX
only the football team was asked	they already prefer football	sample across the school
only pupils in the canteen at lunch	it misses those who bring food	ask from a class list
only five pupils were asked	too few for any conclusion	at least *60*
the question was “isn’t football best?”	the answer is suggested	list the sports neutrally

A LARGE BIASED SAMPLE IS STILL BIASED

Asking *500* people at a football match which sport they prefer is worse than asking *60* pupils chosen properly. Size does not repair a bad sample — only the choosing does.

02 Worked examples

EXAMPLE 1 **Plan a 1-in-10 sample of a school of *600* pupils spread over six grades.**

1. $600 \div 10 = 60$ pupils in the sample.
2. Each grade gives a tenth of itself: $12, 11, 13, 12, 6, 6$.
3. Those add to 60.
   The sample now looks like the school ✓

ANSWER

60 pupils, $12, 11, 13, 12, 6$ and 6 by grade

EXAMPLE 2 **A pupil surveys the football team about the school's favourite sport. Say what is wrong and fix it.**

1. The sample is not representative — every member already prefers football.
2. The conclusion would be true of the team, not of the school.
3. Take every tenth name from the whole school list instead.
   Same effort, honest answer.

ANSWER

A biased sample; sample across the whole school

EXAMPLE 3 **Improve the question “Do you do a lot of sport?”**

1. “A lot” is not defined, so two pupils answer differently for the same behaviour.
2. Ask for a number instead.
3. “How many hours of sport did you do last week?”
   Now the answers can be averaged.

ANSWER

“How many hours of sport did you do last week?”

03 Interactive model

Ask the class the leading question first and count the hands; then ask the neutral one and count again. The gap between the two counts is the lesson.

Planning an honest survey

The whole group being studied is the:

- sample
- population
- tally
- frequency

A 1-in-10 sample of 600 pupils has:

- 6
- 10
- 60
- 600

A grade of 130 contributes:

- 10
- 13
- 30
- 60

Asking only the football team gives a sample that is:

- random
- biased
- systematic
- too large

“Don’t you agree football is best?” is:

- neutral
- leading
- too long
- fine

Age boxes *10–12* and *12–14* are wrong because they:

- overlap
- have gaps
- are too wide
- are too narrow

Taking every tenth name from a list is a sample that is:

- biased
- systematic
- too small
- leading

Asking *500* people at a football match is:

- fine, it is large
- still biased
- random
- systematic

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Investigation	Tadqiqot	Исследование
Population	Bosh to‘plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Representative	Vakillik qiluvchi	Репрезентативный
Bias	Yon bosish	Смещение
Random	Tasodifiy	Случайный
Leading question	Yo‘naltiruvchi savol	Наводящий вопрос
Recording sheet	Qayd varaqasi	Лист записи

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first stage of an investigation is:

- collecting
- planning the question
- drawing a chart
- sampling

A sample should be:

- as small as possible
- representative of the population
- taken from one class
- chosen after the data

A good survey question has:

- open answers
- a small fixed list of answers
- two questions in one
- no options

Bias means the sample:

- is too large
- favours part of the population
- is random
- has no answers

Options must have:

- gaps
- overlaps
- neither gaps nor overlaps
- both

The last stage of the cycle is:

- collecting
- tallying
- interpreting and reporting
- planning

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The whole group being studied is called

   ANSWER The population

2. The part actually asked is called

   ANSWER The sample

3. 60 out of 600 as a fraction

   ANSWER $\frac{1}{10}$

4. …as a percentage

   ANSWER 10%

5. A sample that favours one group is

   ANSWER Biased

6. The first stage of the statistical cycle

   ANSWER Planning the question

7. A question with a fixed list of answers can be

   ANSWER Tallied

Medium · the standard question

7 PROBLEMS

1. A 1-in-10 sample of *600* pupils

   ANSWER *60*

2. Grade 8 has *130* pupils: its share of that sample

   ANSWER *13*

3. Grades of *120*, *110*, *130*, *120*, *60*, *60*: the total

   ANSWER *600*

4. A school of *800* with a 1-in-20 sample

   ANSWER *40*

5. Why ask *60* rather than *6*?

   ANSWER A very small sample is unreliable

6. Asking only the football team is

   ANSWER Biased

7. “Don’t you agree football is best?” is

   ANSWER A leading question

Hard · stretch and reason

7 PROBLEMS

1. A school of *750* with a 1-in-25 sample

   ANSWER *30* pupils

2. *300* boys and *300* girls in a sample of *60*

   ANSWER *30* and *30*

3. *400* boys and *200* girls in a sample of *60*

   ANSWER *40* and *20*

4. Grades of *120* and *60* in a 1-in-10 sample

   ANSWER *12* and *6*

5. Every tenth name from a list of *600* gives

   ANSWER *60* — a systematic sample

6. One reason a lunchtime canteen survey is biased

   ANSWER It misses those who bring food from home

7. The last stage of the cycle

   ANSWER Interpreting and reporting

07 Homework

Homework — 5 tasks

Bring the finished plan to the next lesson: the question, the options, the sample and the sheet.

1. Write your survey question with at most six options, with no gaps and no overlaps.
2. A school has *540* pupils. Plan a 1-in-10 sample and say how many come from a grade of *90*.
3. Explain in one sentence why asking only your own class would bias the result.
4. Rewrite the question “Do you watch a lot of television?” so that it can be averaged.
5. Draw the recording sheet you will use, with a row for every option and a column for the tally.